

FR350SDF Rear Helper Spring Kit Installation Manual

Model: FR350SDF (Rear Suspension Package)

Please read these instructions carefully before starting the installation. Ensure all standard safety precautions are observed while working underneath the vehicle.

1. Parts List

Before beginning, verify that all components listed below are present in your kit.

Ref #	Description / Specifications	Qty
1	Hex Cap Screw (HCS) (3/8" x 1" UNF, Grade 5, Zinc Plated)	2
2	Split Lock Washer (3/8", Grade 5, Zinc Plated)	4
3	Flat Washer USS (3/8" x 1", Grade 2, Zinc Plated)	2
4	Weldment Bracket (For Left-Hand & Right-Hand installation)	2
5	Rubber Spacer (4" O.D. x 1/2" Thick with Bushing)	2
6	Hollow Rubber Spring	2
7	Hex Cap Screw (HCS) (3/8" x 2" UNF, Grade 5, Zinc Plated)	2
8	Hex Cap Screw (HCS) (3/8" x 1-1/2" UNF, Grade 5, Zinc Plated)	2
9	Nylock Locknut (3/8" UNF, Grade 5, Zinc Plated)	2

2. Installation Instructions

- 1. Raise Vehicle:** Elevate the vehicle using an appropriate lifting device, allowing the rear axle to hang freely. Secure the chassis with stands before proceeding.
- 2. Remove OEM Components:** Detach and remove the existing original equipment manufacturer (OEM) bump stop and its associated bushing from the vehicle frame.
- 3. Assemble Helper Spring:** Combine the hollow rubber spring (Ref 6) with the weldment bracket (Ref 4).
- 4. Mount Assembly:** Install the assembled helper spring unit onto the vehicle frame utilizing the existing mounting holes and the provided fasteners (Ref 1, 2, 3, 7, 8, 9).
- 5. Lower Vehicle:** Carefully lower the vehicle back to the ground.

Installation Notes:

- The included 1/2" rubber spacer (Ref 5) is optional. If used, install it with the longer 3/8" x 2" hex cap screw (Ref 7). If the spacer is omitted, use the shorter 3/8" x 1-1/2" hex cap screw (Ref 8).
- Verify that brake lines or cables do not interfere with the helper spring's position or movement.
- When necessary to center the helper spring over the axle, use the 3/8" UNF Nylock locknut (Ref 9) with the offset holes on the bracket.
- If the original equipment vehicle features a peg-style casting at the bump stop location, the helper spring is designed to make contact with this peg. Do not remove the OEM peg.

3. Clearance Instructions & Fitting Guidance

- Clearance:** Depending on vehicle configuration and load, the rubber spring may have a small clearance or light contact when the vehicle is unloaded.
- Ride Comfort:** A small amount of clearance may help maintain a more comfortable unloaded ride. On some Cab & Chassis or heavy-duty applications, light contact or little to no gap may be normal.

- **Spacer Use:** Use the included **1/2" spacer** only when unloaded clearance is noticeably excessive and additional installed height is needed. If there is already a small clearance or light contact, the spacer may not be needed. Avoid excessive preload.
- **Load Effects:** A partial load, cargo, or aftermarket vehicle accessories may further reduce or eliminate clearance.

Important Safety Warnings:

- Do not exceed the vehicle manufacturer's recommended axle weight rating / GAWR.
- Inspect for potential interference between the hollow rubber spring and adjacent vehicle components. Under load, the rubber spring will expand in diameter.

Note: Illustrations are for reference only. Vehicle configuration and actual component appearance may vary. Install according to the vehicle-specific mounting points and standard mechanical practices.